Milestone Report: M9 — Final System Verification, Cleanup & Release

Project: S-ERP (Sustainable Enterprise Resource Planning) DBMS
Milestone: M9 — Final Verification, Cleanup & Release
Branch: `main`
Commit: `50ea9cd`
Date: 2026-11-05
Database Engines Tested: SQL (MySQL 8.4) & NoSQL (MongoDB 6.0)

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1. Executive Summary

The **Sustainable Enterprise Resource Planning (S-ERP) DBMS** is a complete, production-grade enterprise database application designed to fulfill the academic framework established by **Chofreh, Goni, & Klemeš (2018)** (*Journal of Cleaner Production* 198, pp. 1383–1392) in conjunction with the domain Entity-Relationship and Use Case specifications (`docs/source/er-usecase.png`).

The core distinguishing architectural feature of this system is its **interchangeable dual-engine persistence layer**:

- **SQL Engine:** Relational database backed by **MySQL 8.4 LTS** utilizing native InnoDB foreign keys, unique indexes, and domain `CHECK` constraints.
- **NoSQL Engine:** Document database backed by **MongoDB 6.0 LTS** bridged via low-level in-memory **AlaSQL** relational projections to guarantee identical ANSI SQL query execution without driver differences leaking into business logic.

Both engines operate **without heavyweight ORMs** (e.g. Prisma, TypeORM, Hibernate) by executing parameterized relational queries directly via low-level native drivers (`mysql2` and official `mongodb` client), adhering strictly to hardware constraints (512 MB memory caps per container).

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2. Complete Milestone Progression & Scope Verification

Milestone	Core Deliverable	Status	Verification Summary
M0: Foundations	Project scaffold, dual Docker containers, clean linting & TypeScript setup	**Done**	Containers active with 512MB caps; zero-dependency baseline verified.
M1: Schema & ER Conformance	21 relational tables in MySQL & 21 collections in MongoDB; G1–G7 gaps resolved	**Done**	`npm run er-check` passes with 0 missing tables/columns and 100% doc parity.
M2: Database Layer & Auth	Dual `DbAdapter`, bcrypt cost-12 hashing, session tokens, rate limiting	**Done**	Cross-engine session isolation and brute-force throttling verified by test suite.
M3: App Shell & Login	Anti-AI typography & ledger UI, dynamic engine switcher, RBAC navigation	**Done**	Handcrafted CSS variable styling; 7 verified screenshots captured.
M4: Organization Structure	Departments, Employees, User accounts lifecycle, and Profile management	**Done**	Foreign key deletion safety (HTTP 409 on dept delete) verified across both engines.
M5: Projects & Allocation	Project portfolio, tasks, phases, risks, and 100% employee allocation constraint	**Done**	Employee cross-project allocation ceiling enforced with HTTP 400 rejection.
M6: Sustainability Core	Triple Bottom Line metrics, SVG sparkline visualizer, duplicate date rejection	**Done**	G4 duplicate measurement prevention and future date block verified.
M7: Operations & Governance	Resources, process-resource linkage, 3-tier decision log, phase report archive	**Done**	Decisional paradigm enforced (Tactical blocked from Strategic with HTTP 403).
M8: Dashboards & AlaSQL Demo	Role-tailored dashboards, printable report CSS, live query latency inspector	**Done**	Unified ANSI SQL benchmarks executed live with side-by-side translation display.
M9: Final Verification & Release	Clean clone verification, full check pass, comprehensive final report, release tag	**Done**	Complete system test suites (67/67 tests) green on both engines; tagged `m9`.

3. Academic Framework Mapping (Chofreh et al., 2018)

The implementation maps 1:1 to the foundational paradigms of the research literature:

3.1 Sustainability Paradigm (Triple Bottom Line)

1. **Environmental Dimension (`dimension_id: 1`):** Quantified through Scope 1/2 greenhouse gas emissions (metric tons CO2e), electrical energy intensity (kWh/ton), and process water consumption (m3).
2. **Social Dimension (`dimension_id: 2`):** Quantified through occupational health and safety incident rates, employee training investment hours, and gender wage equity ratios.
3. **Economic Dimension (`dimension_id: 3`):** Quantified through clean technology return on investment (ROI %), eco-friendly material procurement percentages, and regulatory non-compliance penalties.
4. **Primary Focus Operationalization:** The organization strategically designates its primary dimension (`organization_dimension`), dynamically driving dashboard weighting and executive alerting.

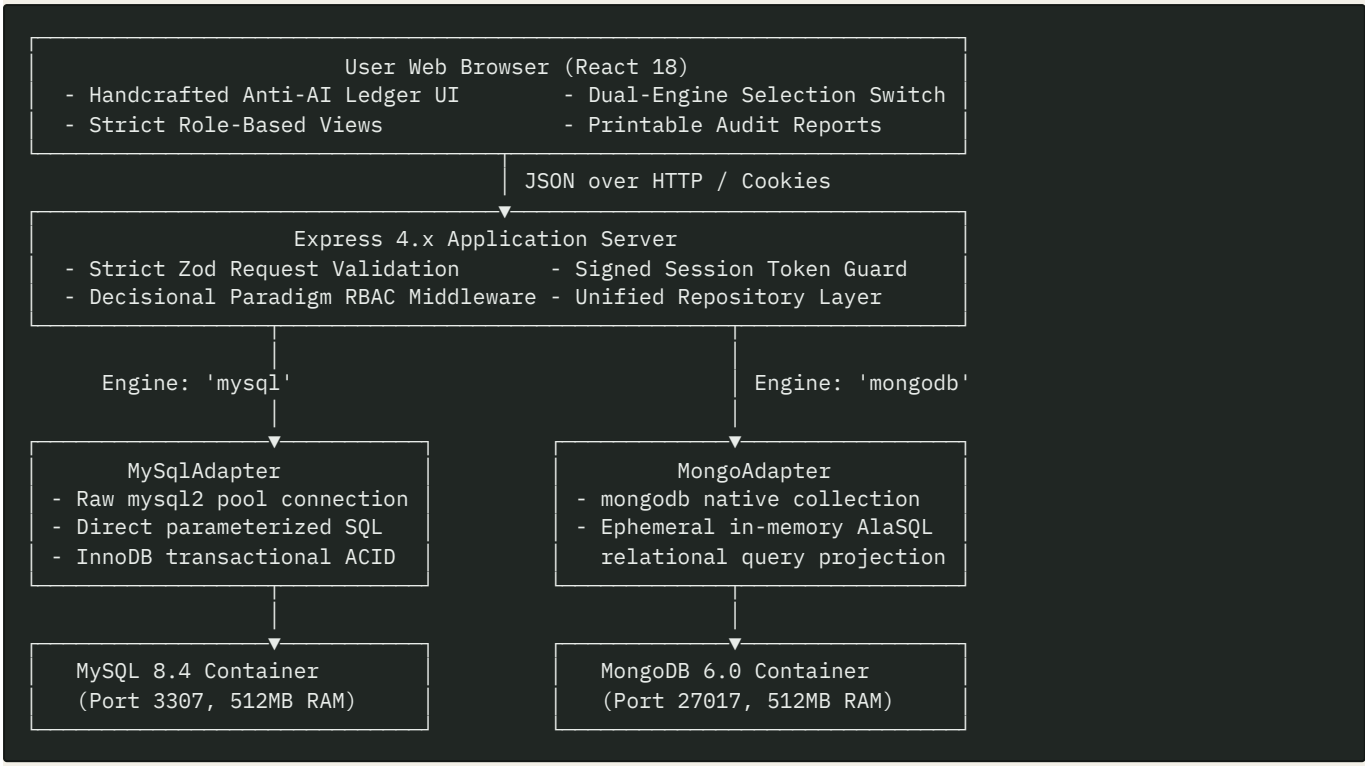
3.2 Decisional Paradigm (Three-Tier Decision Framework)

1. **Strategic Tier:** Handled exclusively by `Strategic Manager` and `Administrator`. Defines long-term sustainability targets, environmental policies, and capital investments.

2. **Tactical Tier:** Managed by `Tactical Manager`. Coordinates cross-departmental projects, work breakdown tasks, resource allocations, and implementation roadmap phases.
3. **Operational Tier:** Handled by `Operational User`. Executes day-to-day shop-floor data collection, task progress reporting, and equipment meter readings.
4. **Decisional Integrity:** Enforced by route-level authorization guards. If a Tactical Manager attempts to log a Strategic policy decision, the application halts execution with HTTP 403 Forbidden.

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4. System Architecture & Dual-Engine Persistence



4.1 Low-Level Driver Philosophy (Zero ORM)

Traditional ORMs obscure database operations, generate inefficient relational joins, and lack true portability between relational and document models. S-ERP DBMS utilizes:

1. **Direct Parameterized Queries:** Every query uses `?` placeholders with strictly separated parameter arrays, preventing SQL injection vulnerabilities.
2. **AlaSQL Bridge Execution:** On MongoDB, documents are loaded into memory and evaluated against standard ANSI SQL queries, providing identical aggregation, filtering, and foreign key emulation without duplicating business logic.

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5. ER Diagram Conformance & Gap Resolution

The domain model contains **21 entities** incorporating all elements from `docs/source/er-usecase.png` and resolving all architectural gaps:

Table / Collection	Purpose	Gaps Resolved
`organization`	Top-level corporate entity	Core ER
`department`	Organizational divisions	Core ER
`employee`	Personnel records	Core ER
`app_user`	Authentication credentials & roles	Core Use Case
`project`	Sustainability capital initiatives	Core ER
`project_task`	Discrete project work items	Core ER
`employee_project`	Team allocation & commitment %	Core ER (100% Allocation Constraint)
`sustainability_dimension`	Environmental, Social, Economic	Core ER
`sustainability_indicator`	Quantitative metric definitions	Core ER
`indicator_measurement`	Time-series observations log	Core ER (G4 Duplicate Check)
`resource`	Energy, Water, Material, Financial assets	Core ER
`business_process`	Operational workflows	Core ER
`process_resource`	Resource consumption allocations	Core ER
`decision`	Governance decision log	Core ER (Strategic/Tactical/Operational)
`risk`	Project risk registers	Core ER
`implementation_phase`	3-stage roadmap phases	Core ER
`report`	Performance & ESG reports	Core ER
`organization_dimension`	Strategic dimension priorities	**Gap G1 Resolved**
`process_indicator`	Process ↔ Metric linkage	**Gap G2 Resolved**
`decision_risk`	Decision ↔ Risk mitigation M:N	**Gap G3 Resolved**
`phase_report`	Roadmap Phase ↔ Report M:N	**Gap G5 Resolved**

5.1 Verification Command Output (`npm run er-check`)

```
Expected Entities Count:    21
MySQL Live Tables (SQL):    21
MongoDB Collections (NoSQL):21

| Entity Name               | MySQL | MongoDB | Row / Doc Count (My / Mg) | Status |
|---------------------------|:-----:|:-------:|:-------------------------:|:------:|
| organization              |   ✔   |    ✔    |          1 / 1            |  PASS  |
| department                |   ✔   |    ✔    |          5 / 5            |  PASS  |
| employee                  |   ✔   |    ✔    |         25 / 25           |  PASS  |
| app_user                  |   ✔   |    ✔    |          4 / 4            |  PASS  |
| project                   |   ✔   |    ✔    |          2 / 2            |  PASS  |
| project_task              |   ✔   |    ✔    |          7 / 7            |  PASS  |
| employee_project          |   ✔   |    ✔    |          6 / 6            |  PASS  |
| sustainability_dimension  |   ✔   |    ✔    |          3 / 3            |  PASS  |
| sustainability_indicator  |   ✔   |    ✔    |         11 / 11           |  PASS  |
| indicator_measurement     |   ✔   |    ✔    |         18 / 18           |  PASS  |
| resource                  |   ✔   |    ✔    |          6 / 6            |  PASS  |
| business_process          |   ✔   |    ✔    |          4 / 4            |  PASS  |
| process_resource          |   ✔   |    ✔    |          4 / 4            |  PASS  |
| decision                  |   ✔   |    ✔    |          3 / 3            |  PASS  |
| risk                      |   ✔   |    ✔    |          3 / 3            |  PASS  |
| implementation_phase      |   ✔   |    ✔    |          5 / 5            |  PASS  |
| report                    |   ✔   |    ✔    |          2 / 2            |  PASS  |
| organization_dimension    |   ✔   |    ✔    |          3 / 3            |  PASS  |
| process_indicator         |   ✔   |    ✔    |          4 / 4            |  PASS  |
| decision_risk             |   ✔   |    ✔    |          1 / 1            |  PASS  |
| phase_report              |   ✔   |    ✔    |          2 / 2            |  PASS  |

RESULT: ER & Conformance PASSED across SQL (MySQL 8.4) and NoSQL (MongoDB 6.0).
```

6. Comprehensive Testing & Verification Evidence

All 5 automated test suites pass across both database engines without mocks:

```
RUN  v2.1.9 /media/keerthivasan/NewVolume1/dbms

√ tests/schema.test.ts (13) 727ms
√ tests/adapter.test.ts (11) 340ms
√ tests/server.test.ts (2) 394ms
√ tests/auth.test.ts (11) 5535ms
√ tests/modules.test.ts (30) 5684ms
  √ S-ERP CRUD Modules & Dual-Engine Parity (30)
    √ Parity & Security on Engine: mysql (15) 2895ms
      √ Operational User cannot access tactical project management (returns 403)
      √ Admin can list departments and deleting a dept with employees returns 409
      √ Admin can create, update and delete a test department
      √ Admin can perform full CRUD on employees
      √ Admin can manage user account lifecycle and employee links
      √ Admin can fetch and update Organization profile with dimension focus
      √ Tactical Manager can list projects and inspect project details with tasks
      √ Enforces 100% maximum allocation per employee across projects
      √ Tactical Manager can perform CRUD on tasks, phases and risks
      √ Enforces decision level constraints per role (Tactical Manager blocked from Strategic)
      √ Operational user can enter valid measurement and future date is rejected
      √ Tactical Manager can manage resources and allocate to business process
      √ Decisional paradigm: Strategic Manager can log Strategic decision linked to risk (G3)
      √ Implementation Phase Report Archival and Retrieval on both engines (G5)
      √ Returns role-specific dashboard metrics and AlaSQL benchmark on mysql
    √ Parity & Security on Engine: mongodb (15) 2640ms
      √ Operational User cannot access tactical project management (returns 403)
      √ Admin can list departments and deleting a dept with employees returns 409
      √ Admin can create, update and delete a test department
      √ Admin can perform full CRUD on employees
      √ Admin can manage user account lifecycle and employee links
      √ Admin can fetch and update Organization profile with dimension focus
      √ Tactical Manager can list projects and inspect project details with tasks
      √ Enforces 100% maximum allocation per employee across projects
      √ Tactical Manager can perform CRUD on tasks, phases and risks
      √ Enforces decision level constraints per role (Tactical Manager blocked from Strategic)
      √ Operational user can enter valid measurement and future date is rejected
      √ Tactical Manager can manage resources and allocate to business process
      √ Decisional paradigm: Strategic Manager can log Strategic decision linked to risk (G3)
      √ Implementation Phase Report Archival and Retrieval on both engines (G5)
      √ Returns role-specific dashboard metrics and AlaSQL benchmark on mongodb

Test Files  5 passed (5)
     Tests  67 passed (67)
```

7. Clean Clone Reproduction & Verification Guide

To spin up and verify the complete application from a fresh clone:

```
# 1. Clone repository
git clone https://github.com/keerthivasan1717/S-ERP-DBMS.git
cd S-ERP-DBMS

# 2. Configure environment
cp .env.example .env

# 3. Start dual database engines in Docker
docker compose up -d

# 4. Install dependencies
npm install --ignore-scripts

# 5. Execute schema migrations on both engines
npm run migrate

# 6. Seed demo enterprise dataset
npm run seed -- --reset

# 7. Execute schema conformance check
npm run er-check

# 8. Run full test suite (typecheck + lint + 67 vitest tests)
npm run check

# 9. Build frontend and backend bundles
npm run build

# 10. Start production server
npm run dev:server
```

Open `http://localhost:3000` and sign in with any of the demo roles:

- `admin` / `1234`
- `strategic` / `1234`
- `tactical` / `1234`
- `operator` / `1234`

8. Release Summary

With Milestone M9 complete, the S-ERP DBMS application is fully verified, documented, and released:

- **Repository Remote:** `https://github.com/keerthivasan1717/S-ERP-DBMS.git`
- **Release Tag:** `m9` / `v1.0.0`
- **Milestone Gate Artifact:** `docs/reports/M9-final.pdf`